Where Voice Becomes Measurable Regulation By Translating Acoustic Stability Into Actionable Health Intelligence

The Adaptive Regulation Intelligence Platform

Cross-Sector Application Framework

Breathermae's Voice-Based Neuro-Autonomic Regulation Engine is designed as a universal adaptive capacity signal, not as a tool confined to a single population or use case. At its core, it measures how effectively the human nervous system adapts under load whether that load arises from athletic competition, occupational demand, community stressors, environmental exposure, or the physiological shifts associated with aging. Across all contexts, the underlying biological question remains the same: how stable, flexible, and resilient is the system when challenged?

Rather than diagnosing conditions or labeling emotional states, this platform translates autonomic regulation patterns into structured intelligence that integrates seamlessly with the Art of Wellness, BSI™ Systems Intelligence, and 360° Health Intelligence. Subjective perception is paired with objective regulation data, allowing Load, Capacity, and Strain to be modeled in real time and over time. The result is a scalable architecture that functions equally well for elite performers, corporate leaders, municipalities, Blue Zone communities, and retirement populations.

What differs across these verticals is not the signal itself, but the context in which it is interpreted. For athletes, the focus is readiness and recovery velocity. For corporate environments, it is resilience and burnout prevention. For communities, it is population-level stress adaptation. For aging populations, it is durability and preserved flexibility. The biological principle remains constant: adaptive regulation capacity determines performance, resilience, and longevity.

By measuring adaptation before breakdown occurs, Breathermae provides a unifying regulation intelligence layer that scales from the individual to the enterprise to the community. This is not a mental health tool, nor a disease detection system. It is an adaptive capacity platform designed to strengthen resilience across the full human spectrum from high performance to healthy aging.

What Makes This Engine Fundamentally Different

Most voice technologies on the market today are built for speech recognition, sentiment analysis, or behavioral profiling. Platforms such as Google Speech-to-Text, Amazon Alexa Voice Services, Apple Siri, Microsoft Azure Cognitive Services, and OpenAI Whisper are optimized to transcribe words, detect intent, classify tone, or estimate emotional likelihood. These systems interpret language and content. They are designed to understand what is being said.

Breathermae’s architecture operates in an entirely different category. It does not analyze vocabulary, emotional labeling, psychological inference, or conversational intent. It does not attempt to diagnose mental states or assign clinical meaning. Instead, it converts standardized voice samples into frequency-based regulatory signatures that model autonomic balance, tension distribution, neuromuscular coordination, and adaptive variability over time.

Where traditional systems interpret language, this engine models regulation. Integrated within the Breathermae Systems Intelligence ecosystem, the engine functions as a non-invasive regulatory biomarker — translating acoustic stability into measurable load, capacity, coordination, and resilience metrics. It bridges subjective experience with physiologic pattern modeling inside the BSI™ framework, strengthening proactive health management without crossing into medical diagnosis.

This is not sentiment analysis. This is regulation intelligence.

Strategic Relevance to Breathermae

THIS BECOMES:

- A non-invasive neural load sensor
- A daily check-in tool
- A corporate resilience tracker
- A performer readiness signal
- A PTSD-support regulation monitor (without diagnosing)

IT FITS INSIDE:

Art of Wellness → subjective
Voice Engine → autonomic
BSI → integrative interpretation

TOGETHER:

You measure how someone feels
You measure how someone is adapting
You measure how systems are coordinating

That is the differentiation.

Athlete & Performer Version (BSI-AP)

This Becomes:

• A non-invasive Autonomic Load Monitor
• A daily neural regulation readiness check
• A training strain calibration signal
• A recovery efficiency indicator
• A performance stability tracker
• A regulation support tool under competitive stress

Without diagnosing. Without labeling. Pure signal intelligence.

It Integrates Directly Into:

Art of Wellness → Subjective fatigue, mood, stress perception
Voice Regulation Engine → Objective neural-autonomic load
BSI-AP (Athlete Performance) → Load, Capacity, Strain modeling
360° Health Intelligence → Trend analytics + physiological correlation

This creates a closed-loop performance system.

What It Allows Coaches to See:

• When nervous system load is rising before performance drops
• When recovery speed is slowing after high-output events
• When coordination stability narrows under pressure
• When adaptive capacity is expanding through training

Not emotion detection.
Not psychological profiling.
Regulation dynamics.

For the Athlete / Performer:

You measure readiness.
You measure recovery.
You measure resilience under load.

That protects performance longevity.
That reduces overtraining risk.
That is BSI-AP differentiation.

Bull Rider Performance Version (BSI-AP Rodeo)

This Becomes:

- A non-invasive Nervous System Readiness Signal
- A pre-ride regulation stability check
- A post-ride recovery velocity indicator
- A high-adrenaline load tracking tool
- A resilience monitor across rodeo weekends
- A long-term neural durability metric

Without diagnosing.
Without labeling.
Pure autonomic regulation intelligence.

Why This Matters in Bull Riding

Bull riding is not endurance.
It is explosive autonomic control under chaos.

The difference between an 88-point ride and a buck-off can be:

- Microsecond reaction timing
- Vestibular stabilization
- Grip coordination under sympathetic surge
- Recovery speed after adrenal spikes
- Emotional regulation after a hard hit

This engine measures regulation under pressure — not just strength or conditioning.

It Integrates Into:

Art of Wellness → Subjective fatigue, focus, mental edge
Voice Regulation Engine → Objective autonomic load
BSI-AP Rodeo Model → Load vs Capacity vs Strain
Performance Dashboard → Coach-visible trend intelligence

Now coaches see what the athlete cannot feel yet.

What It Allows Coaches to Track:

- Pre-competition regulation stability
- Weekend cumulative stress load
- Recovery speed between rides
- Autonomic overactivation trends
- Narrowing coordination windows under pressure
- Early signals of overtraining or burnout

Not emotion detection.
Not psychological assessment.
Regulation capacity under extreme stress.

For the Rider:

You measure how ready you are.
You measure how fast you recover.
You measure how steady your system stays when everything is moving.

That protects:

- Ride consistency
- Injury risk
- Cognitive sharpness
- Career longevity

That is BSI-AP Rodeo differentiation.

Corporate Wellness & Workforce Resilience Version

This Becomes:

• A non-invasive Workforce Regulation Stability Indicator
• A daily executive stress calibration tool
• A resilience and burnout risk trend tracker
• A leadership performance stability signal
• A corporate load vs capacity analytics layer
• A proactive nervous system support monitor — without diagnosing

Not mental health screening.
Not psychological labeling.
Regulation intelligence at scale.

Why This Matters in Bull Riding

High-performing organizations operate under sustained cognitive and emotional load.

The cost of dysregulation shows up as:

• Reduced decision quality
• Increased absenteeism
• Burnout and turnover
• Leadership volatility
• Decreased cognitive flexibility
• Team coordination breakdown

This engine measures how the nervous system is adapting to stress before productivity declines.

It Integrates Into:

Art of Wellness → Subjective perception of stress and work-life strain
Voice Regulation Engine → Objective autonomic regulation signal
BSI™ Systems Intelligence → Load, Capacity, Strain modeling
Corporate Analytics Dashboard → Aggregated, anonymized resilience trends

Individual privacy preserved.
Population-level intelligence enabled.

What It Allows Organizations to See:

• Workforce regulation stability trends
• Departments with narrowing resilience windows
• Recovery speed after high-pressure cycles
• Cumulative stress load across quarters
• Burnout risk signals before HR metrics shift
• Impact of corporate initiatives on physiological resilience

This becomes a Preventive Health Momentum Score at the enterprise level.

For Leadership:

You measure how your people feel.
You measure how they are adapting.
You measure how systems are coordinating under load.

That Reduces:

• Burnout costs
• Health-related absenteeism
• Executive turnover
• Insurance strain

That Strengthens:

• Performance consistency
• Cognitive clarity
• Team resilience
• Long-term workforce durability

That is Breathermae's corporate differentiation.

Communities & Blue Zones Version

This Becomes:

- A Community Regulation Stability Indicator
- A Population-Level Resilience Signal
- A Preventive Health Momentum Tracker
- A Stress Adaptation Trend Monitor
- A Blue Zone Alignment Index Contributor
- A Non-diagnostic Nervous System Health Barometer

Not disease screening.
Not mental health labeling.
Community resilience intelligence.

Why This Matters for Blue Zones & Municipalities

Communities influence:

- Stress exposure
- Social cohesion
- Environmental load
- Lifestyle alignment
- Access to restorative spaces
- Economic pressure

Yet most cities measure health only through:

- Hospital utilization
- Chronic disease prevalence
- Insurance data
- Mortality statistics

That is reactive.

This engine measures adaptive capacity before disease manifests.

It Integrates Into:

Art of Wellness (8 Pillars)
→ Physical, Mental, Emotional, Social, Environmental, Financial insights

Voice Regulation Engine
→ Objective autonomic adaptation signal

BSI™ Systems Intelligence
→ Community Load, Capacity, and Coordination modeling

Community Resilience Dashboard
→ Anonymous, aggregated population trends

Individual data remains private.
City-level patterns become visible.

What Municipal Leaders Can See:

- Shifts in community stress regulation over time
- Impact of infrastructure improvements
- Effects of green spaces and walkability
- Economic stress signals before crisis
- Resilience differences between neighborhoods
- Recovery trends after environmental or social disruptions

This becomes a Community Resilience Index grounded in real human adaptation — not just statistics.

For Blue Zone Alignment:

You measure:

- How people feel
- How they are adapting
- How community systems are coordinating

You correlate that with:

- Walkability
- Social connectedness
- Access to whole foods
- Environmental exposures
- Economic opportunity

That is proactive health intelligence at scale.

The Differentiation:

Hospitals measure illness.
Cities measure utilization.
Breathermae measures adaptation.

That is how a community becomes resilient before it becomes reactive.

Aging Population & Retirement Communities Version

This Becomes:

• A Nervous System Stability Indicator for Aging Adults
• A Recovery Preservation Signal
• A Cognitive Resilience Trend Tracker
• A Social Isolation Stress Monitor
• A Fall-Risk Correlation Support Layer
• A Longevity Adaptation Index Contributor

Not disease diagnosis.
Not neurological screening.
Not dementia detection.

A regulation durability signal.

Why This Matters in Aging Populations

Aging is not defined only by years lived.
It is defined by adaptive flexibility retained.

As we age, the nervous system can:

• Recover more slowly
• Narrow its variability range
• Lose coordination efficiency
• Become more stress-reactive
• Show reduced autonomic flexibility

These shifts often precede:

• Cognitive decline
• Mobility instability
• Sleep disruption
• Social withdrawal
• Increased healthcare utilization

The key is not diagnosing disease.

The key is measuring adaptive capacity over time.

It Integrates Into:

Art of Wellness (8 Pillars)
→ Social engagement, purpose, physical vitality, emotional balance

Voice Regulation Engine
→ Objective autonomic regulation stability

BSI™ Systems Intelligence
→ Load vs Capacity modeling in aging physiology

Retirement Community Dashboard
→ Anonymous trend tracking across residents

Privacy preserved.
Population trends visible.

What Retirement Communities Can Monitor:

• Regulation stability across months and years
• Slowing recovery velocity
• Narrowing variability windows
• Social isolation stress patterns
• Impact of movement programs
• Impact of community engagement activities

This allows proactive programming — before decline accelerates.

For the Aging Individual:

You measure:

• How steady your system remains
• How quickly you recover
• How adaptable you stay

Longevity is not only lifespan.

It is preserved regulation capacity.

Why This Is Blue Zone Aligned:

Blue Zones emphasize:

• Movement
• Social cohesion
• Purpose
• Environmental alignment
• Stress modulation

This engine provides a measurable regulation layer that correlates directly with those factors.

It allows cities and retirement communities to measure whether their initiatives are strengthening adaptive resilience.

Aging Population & Retirement Communities

The Broader Integration

Athletes measure readiness.
Corporations measure resilience.
Communities measure vitality.
Aging populations measure durability.

Same signal.
Different application horizon.

That Is the Unified Breathermae Architecture

How It Works

Voice-Based Neuro-Autonomic Regulation Analysis Engine

At its core, the system analyzes how the nervous system regulates under load by examining structural properties of the human voice.

- It does not analyze words.
- It does not interpret emotional meaning.
- It does not diagnose conditions.

Instead, it measures regulation dynamics embedded within vocal acoustics.

1 Standardized Voice Capture

A short, structured voice sample is collected using a guided protocol.

The script is content-neutral and designed to produce consistent phonation across individuals and timepoints.

The goal is not what is said — but how the voice stabilizes, transitions, and sustains under controlled conditions.

This creates a repeatable regulatory snapshot.

2 Signal Conditioning

The raw waveform undergoes preprocessing:

- Environmental noise reduction
- Amplitude normalization
- Device compensation
- Phonation separation from breath artifacts

This ensures that the output reflects biological regulation patterns — not recording variability.

3 Frequency Decomposition

The conditioned signal is converted from the time domain into frequency-domain representation.

The system analyzes:

- Energy distribution across configurable frequency bands
- Harmonic layering
- Spectral density
- Stability of tonal structures
- Micro-instability fluctuations

Unlike musical pitch analysis, this does not assign emotional meaning to frequencies.
It evaluates structural regulation.

4 Feature Extraction & Pattern Modeling

From the decomposed signal, the engine extracts measurable regulation markers:

- Variability indices
- Harmonic coherence
- Transitional smoothness
- Micro tremor rate
- Noise-to-tone ratio
- Symmetry distribution
- Stability under sustained phonation

These markers serve as proxies for autonomic coordination and neuromuscular tension patterns.

5 Autonomic Regulation Modeling

The extracted features are mapped into a non-diagnostic regulatory model that estimates:

- Sympathetic activation patterns
- Parasympathetic engagement
- Regulatory rigidity vs flexibility
- Recovery bandwidth
- Adaptive transition capacity

Importantly, the system models balance and coordination not emotional state.

6 Baseline & Longitudinal Tracking

Each individual establishes a personal baseline.

Future samples are compared against:

- Prior regulatory state
- Recovery slope
- Load accumulation
- Adaptation improvement

Over time, this creates a trajectory of resilience and strain patterns.

This is where the defensibility lies.

7 Systems Intelligence Integration

Outputs are integrated into:

- BSI™ strain and capacity scoring
- Wearable metrics (HRV, GSR, etc.)
- Sleep and recovery data
- Self-reported wellness inputs
- Biomarker panels

Voice becomes one regulatory input layer within a multi-dimensional intelligence model.

Not standalone.
Not diagnostic.
Integrated.

What Makes This Different

Most voice systems ask:

"What is this person feeling?"

Breathermae asks:

"How effectively is this nervous system regulating?"

That is a fundamentally different question.

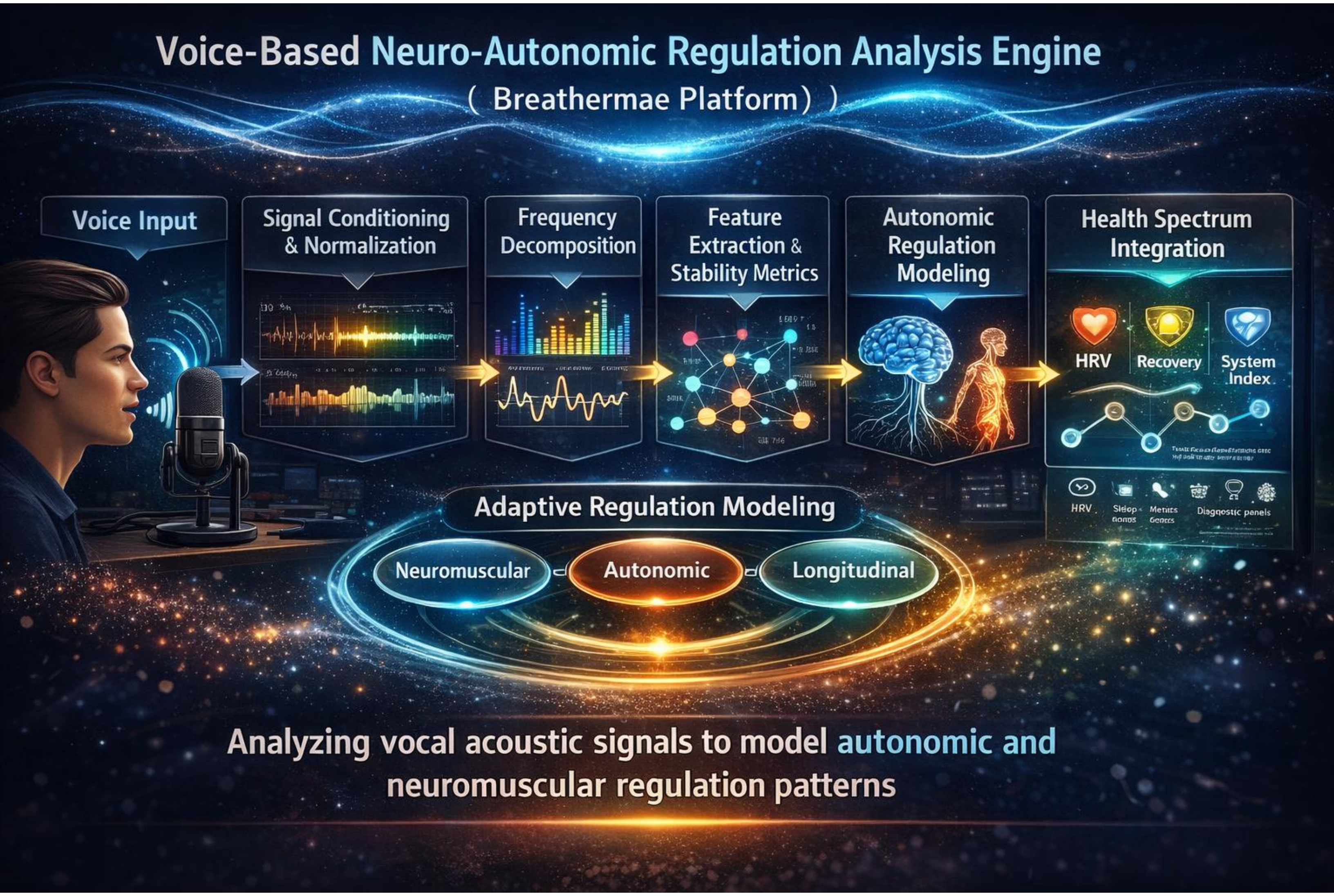


Figure 1: This figure illustrates the staged architecture of the Breathermae Voice-Based Neuro-Autonomic Regulation Analysis Engine. It depicts the sequential pipeline from voice input and signal conditioning through frequency decomposition and feature extraction, into autonomic regulation modeling and health spectrum integration. The framework demonstrates how vocal acoustic signals are transformed into neuromuscular and autonomic regulation metrics that feed adaptive, longitudinal modeling within the Breathermae platform.

Technical Implementation Blueprint & Systems Integration Architecture

STAGE 0 — System Positioning Inside Breathermae

Before engineering begins, this must be clear internally:

This engine is:

- A signal-derived regulation index
- A neural-autonomic load indicator
- A BSI™ Load / Capacity contributor
- NOT a diagnostic emotional classifier
- NOT a mental health diagnostic tool
- NOT a frequency = emotion mapping system

It feeds into:

- Neural-Emotional Load (BSI domain)
- Adaptive Capacity
- Stress Regulation Trend
- Longitudinal Regulation Stability Index

STAGE 1 — Controlled Voice Capture Layer

This is where your biomedical engineer starts.

Hardware-Agnostic Capture Layer

- 16-bit minimum resolution
- 44.1 kHz sampling rate (minimum)
- Consistent microphone gain normalization
- Ambient noise threshold detection
- RMS stabilization pre-record

Capture Protocol

User performs:

1. Sustained vowel (e.g., “Ahhh”) – 8 seconds
2. Controlled breath-paced reading – 20 seconds
3. Neutral phrase repetition – 3 repetitions

Purpose:

- Capture steady-state phonation
- Capture cognitive-motor speech coordination
- Capture respiratory-laryngeal synchronization

Output of Stage 1:

Raw WAV file

Metadata:

- Timestamp
- Device type
- Mic baseline amplitude
- Noise floor
- User ID (tokenized)

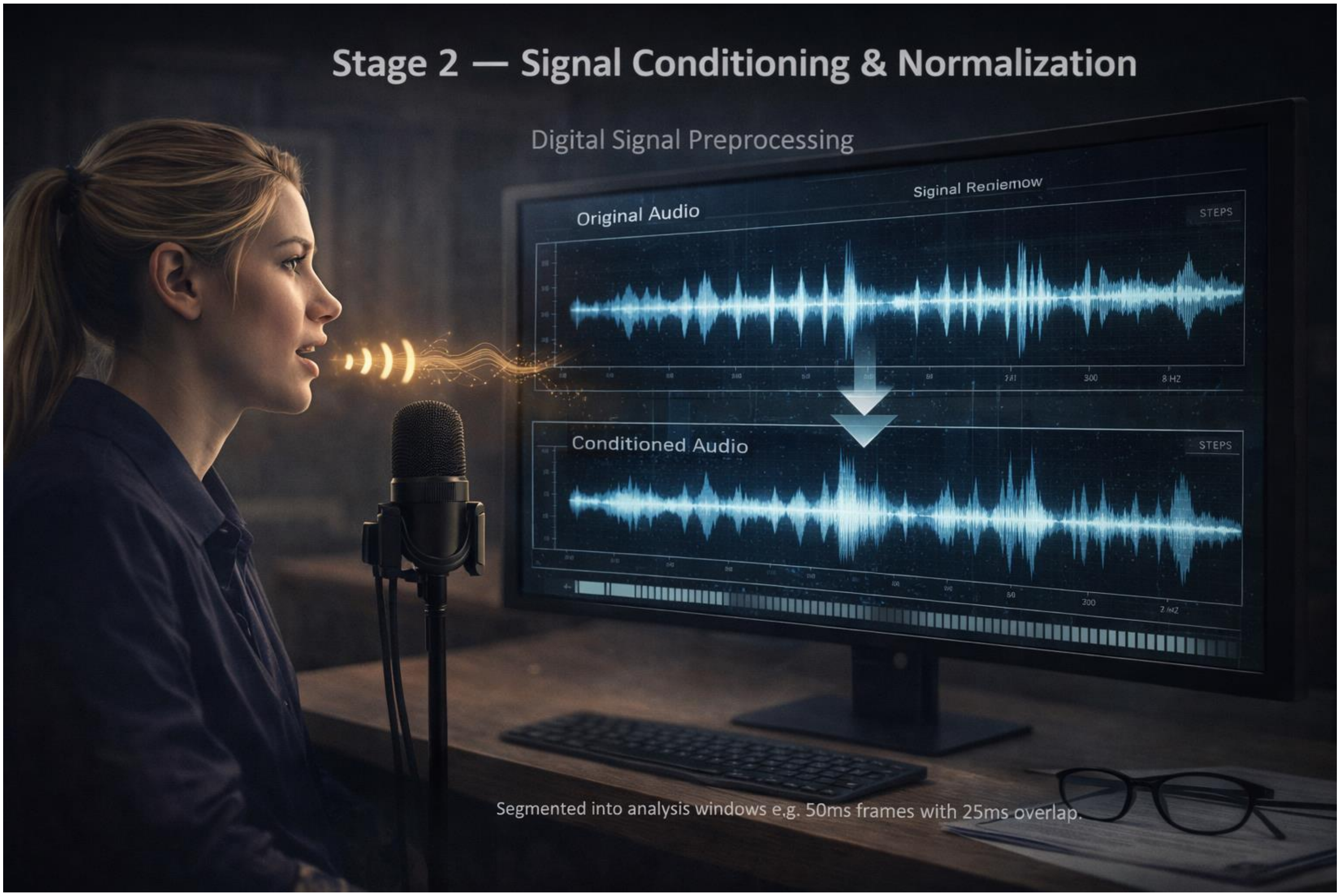
This stage is purely acquisition.

What your run produced:
•results/stage1_wavs/
•results/stage1_metadata_qc.csv
•results/stage1_results.json
That means Stage 1 completed as an acquisition and QC pass, which is exactly the intended purpose of this stage.

The code aligns with these Stage 1 elements:
- Controlled Voice Capture Layer
- Hardware-agnostic capture
- 16-bit minimum resolution through ffmpeg conversion
- 44.1 kHz minimum sample rate
- Consistent microphone gain normalization readiness
- Ambient noise threshold detection
- RMS stabilization of the pre-record baseline
- Metadata capture including timestamp, device type, mic baseline amplitude, noise floor, and tokenized user ID
- Outcome framing: “This stage is purely acquisition”

Your current Stage 1 results indicates that:
- the raw `.m4a` recordings were successfully converted into usable `.wav` files
- basic signal-quality and acquisition checks were completed successfully
- the pipeline generated a metadata QC file for backend use and a structured JSON results file for readable output and later stages
- Stage 1 is functioning correctly as an acquisition and quality-control layer rather than a diagnostic or interpretation layer

That is the correct role of Stage 1.



STAGE 2 — Signal Conditioning & Normalization

This is digital preprocessing.

Processing Steps

1. DC offset removal
2. Band-pass filter (80 Hz – 8 kHz typical speech band)
3. Noise suppression via spectral subtraction
4. Amplitude normalization
5. Silence trimming

Output:

Clean waveform buffer

Segmented into analysis windows (e.g., 50ms frames with 25ms overlap)

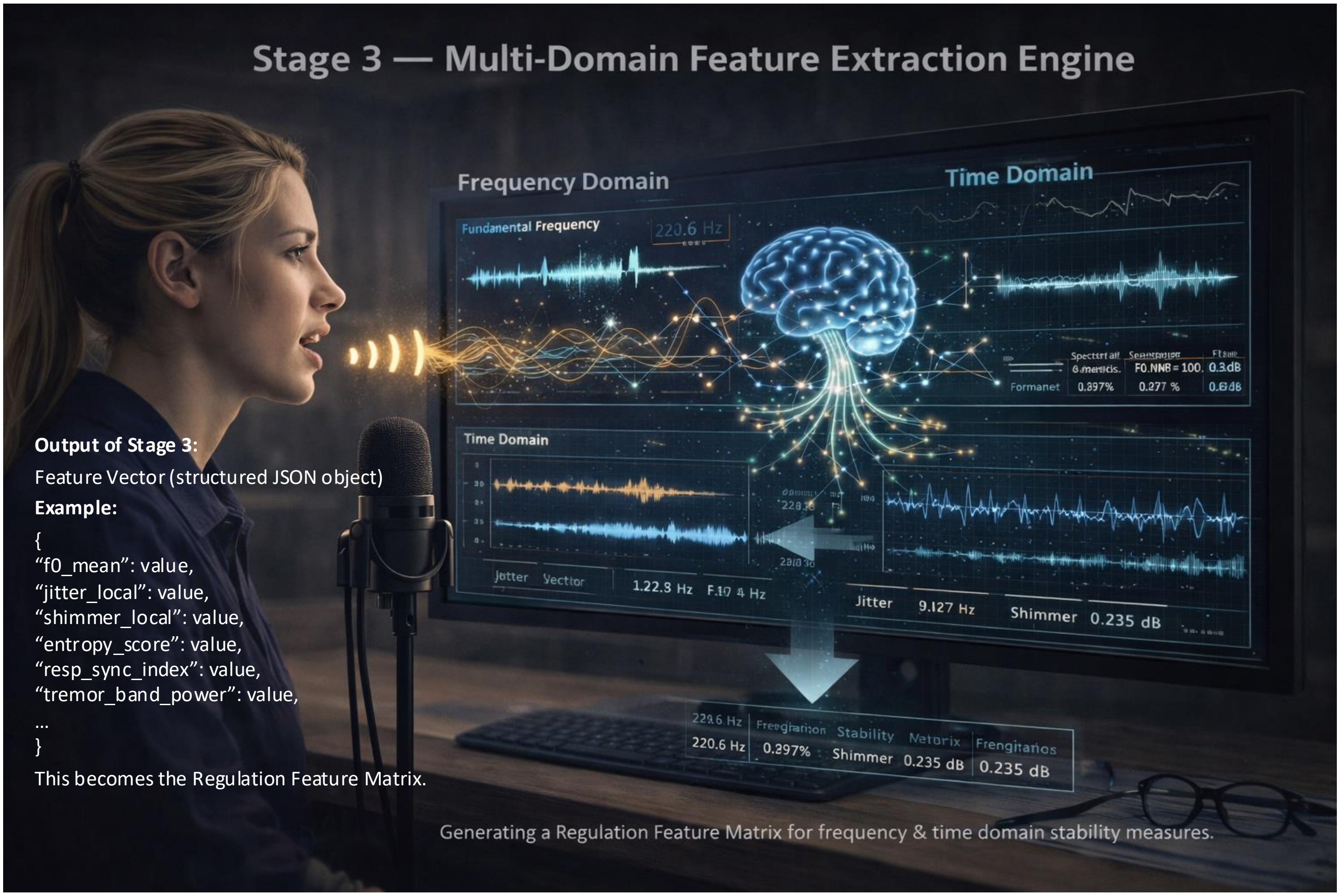
What your run produced:
- results/stage2_wavs/
- results/stage2_metadata_qc.csv
- results/stage2_results.json

That means Stage 2 completed as a **signal conditioning and preprocessing pass**, which is exactly the intended purpose of this stage.

The code aligns with these Stage 2 elements:
- Signal Conditioning & Normalization
- DC offset removal
- Band-pass filtering (80 Hz – 8 kHz speech band)
- Environmental noise reduction via spectral subtraction
- Amplitude normalization (standardized signal intensity)
- Device compensation (adjustment for microphone variability)
- Silence trimming (removal of non-informative segments)
- Breath / phonation separation via windowing
- Output segmentation into analysis windows (50 ms frames with 25 ms overlap)
- Outcome framing: "This stage is digital preprocessing"

Your current Stage 2 results indicate that:
- the Stage 1 WAV files were successfully conditioned into clean, analysis-ready signals
- background noise and low-frequency / high-frequency interference were reduced
- signal intensity was standardized across recordings, reducing device-related variability
- silence and non-informative portions of the signal were removed
- the waveform was segmented into consistent analysis windows for downstream feature extraction
- the pipeline generated a metadata QC file for backend validation and a structured JSON results file for readable output and later stages
- Stage 2 is functioning correctly as a preprocessing and normalization layer rather than a feature extraction or interpretation layer

That is the correct role of Stage 2.



STAGE 3 — Multi-Domain Feature Extraction Engine

This is where engineering becomes serious.

You are not extracting emotion.

You are extracting regulation stability metrics.

A. Frequency Domain

- Fundamental frequency (F0)
- Harmonic-to-noise ratio
- Spectral centroid
- Spectral bandwidth
- Formant distribution (F1–F3)

Purpose:

Assess neuromuscular micro-instability.

B. Time Domain

- Jitter (cycle-to-cycle frequency variation)
- Shimmer (amplitude instability)
- Zero crossing rate
- Envelope variance
- Tremor detection bands (3–12 Hz modulation)

Purpose:

Micro-motor regulation stability.

C. Nonlinear Dynamics

- Approximate entropy
- Sample entropy
- Fractal dimension
- Lyapunov exponent (short-window estimate)

Purpose:

Autonomic signal irregularity index.

D. Respiratory-Coupling Metrics

Extract amplitude modulation tied to breath cycles.

Measure:

- Breath-speech synchronization lag
- Respiratory amplitude variance
- Phonation stability during exhalation

Purpose:

Autonomic-respiratory coordination integrity.

What your run produced:

- results/stage3_feature_vectors.csv
- results/stage3_results.json

That means Stage 3 completed as a **multi-domain feature extraction pass**, which is exactly the intended purpose of this stage.

The code aligns with these Stage 3 elements:

- Multi-Domain Feature Extraction Engine
- Frequency decomposition from sound to measurable structure
- Domain conversion from time domain to frequency domain
- Energy distribution analysis across signal bands
- Structural acoustic analysis
- Fundamental frequency (F0)
- Harmonic-to-noise ratio
- Spectral centroid
- Spectral bandwidth
- Formant distribution (F1–F3)
- Time-domain micro-instability metrics
- Jitter
- Shimmer
- Zero crossing rate
- Envelope variance
- Tremor band power (3–12 Hz modulation)
- Nonlinear dynamics metrics
- Approximate entropy
- Sample entropy
- Fractal dimension
- Lyapunov-style short-window instability proxy
- Respiratory-coupling metrics
- Breath-speech synchronization lag
- Respiratory amplitude variance
- Phonation stability during exhalation
- Analytical positioning that does not assign emotional meaning
- Outcome framing: extraction of **regulation stability metrics**

Your current Stage 3 results indicate that:

- the conditioned audio signals were successfully transformed into a structured **regulation feature matrix**
- the voice signal has been decomposed into both frequency-domain and time-domain representations, enabling detailed structural analysis
- tonal structure and harmonic organization were quantified through frequency-based features such as F0, spectral centroid, and bandwidth
- micro-instability within the signal was captured through jitter, shimmer, and envelope-based variability measures
- nonlinear complexity and irregularity were quantified using entropy-based and fractal-style metrics
- tremor-related modulation energy within the 3–12 Hz band was successfully isolated and measured
- respiratory-phonatory coordination was quantified through synchronization lag, amplitude variance, and exhalation stability metrics
- the pipeline generated a feature vector CSV for backend modeling and a structured JSON results file for readable output and downstream stages
- Stage 3 is functioning correctly as a **feature extraction layer**, converting raw signal into measurable physiological proxies rather than performing interpretation or diagnosis

That is the correct role of Stage 3.

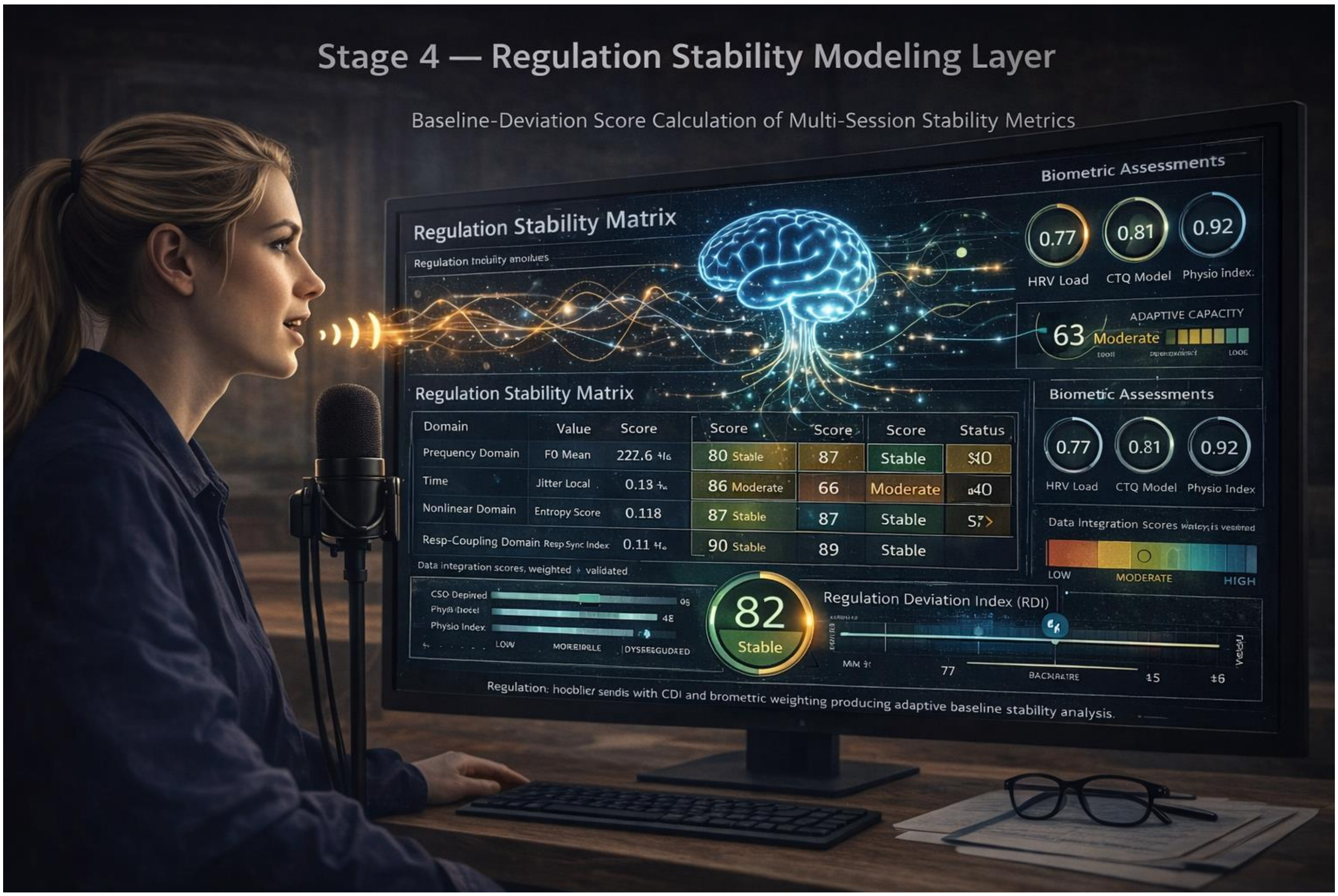
Output of Stage 3:

Feature Vector (structured JSON object)

Example:

{
“f0_mean”: value,
“jitter_local”: value,
“shimmer_local”: value,
“entropy_score”: value,
“resp_sync_index”: value,
“tremor_band_power”: value,
…
}

This becomes the Regulation Feature Matrix.



STAGE 4 — Regulation Stability Modeling Layer

This is the core differentiator.

You are not mapping to emotions.

You are calculating:

- Stability
- Variability
- Coordination
- Deviation from personal baseline

Step 4A — Baseline Establishment

First 3 sessions establish:

- Personal harmonic baseline
- Variability tolerance band
- Regulation window

This is user-specific modeling.

What your run produced:

- results/stage4_results.json
- results/stage4_baseline_history.json

That means Stage 4 completed as a **regulation stability modeling and baseline-tracking pass**, which is exactly the intended purpose of this stage.

The code aligns with these Stage 4 elements:

- Regulation Stability Modeling Layer
- Variability indices (measuring fluctuation patterns in the signal)
- Harmonic coherence (assessing alignment and structure of harmonics)
- Transitional smoothness (evaluating continuity between vocal states)
- Micro-tremor detection (capturing fine-grain instability patterns)
- Noise-to-tone ratio (comparing signal clarity vs noise components)
- Symmetry distribution (measuring balance across signal structure)
- Sustained phonation stability (assessing consistency over time)
- Baseline establishment (tracking session history across runs)
- Deviation index calculation (relative to personal baseline)
- Analytical positioning that does not assign emotional meaning
- Outcome framing: extraction of **regulation stability indices**

Your current Stage 4 results indicate that:

- the Stage 3 feature vectors were successfully transformed into higher-level **regulation stability metrics**
- domain-level scores were computed representing stability, variability, coordination, and structural integrity
- a session-level record was stored in the **baseline history file** for future comparison
- baseline tracking has begun across sessions, enabling future deviation modeling
- with fewer than 3 sessions, the system is operating in a **provisional baseline mode**
- deviation scoring logic is initialized but not yet fully statistically stable
- the pipeline generated a structured JSON results file for readable output and downstream modeling
- Stage 4 is functioning correctly as a **modeling and aggregation layer**, not a preprocessing or feature extraction stage

<u>IMPORTANT NOTE:</u>

- A **true personal baseline requires 3 separate sessions** (each with the full set of recordings)
- Your current results are **provisional** and only represent **initial baseline accumulation**
- Until 3 sessions are collected, **deviation scores and RDI are not fully reliable**
- You will need to **run the same code 3 different times**, each with a new session of audio files
- Each run will **automatically append to the baseline history file** (no code changes needed)
- After 3 sessions, the code will begin **true baseline comparison and stable deviation modeling**
- Additional sessions beyond 3 will **improve accuracy, tolerance bands, and long-term tracking**

Step 4B — Deviation Index Calculation

Each session computes:

Regulation Deviation Index (RDI)

RDI = Weighted instability composite relative to baseline

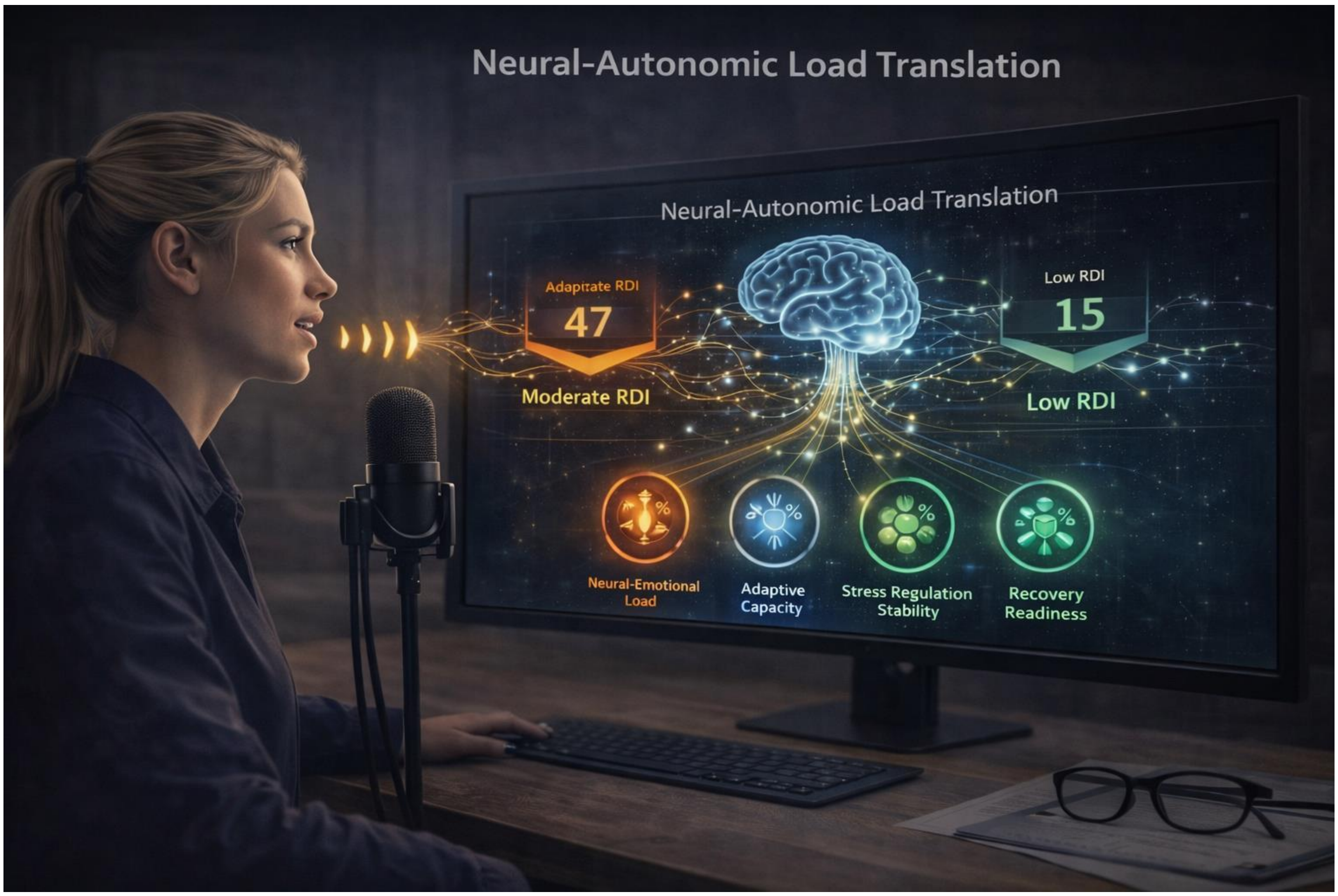
Weights determined by:

- CSO-defined physiologic significance
- CTO normalization modeling
- Biomedical validation thresholds

Output:

RDI Score (0–100 scale)

Lower = stable
Higher = dysregulated variability



STAGE 5 — Neural-Autonomic Load Translation Engine

Now it integrates into BSI™.

Mapping logic (non-diagnostic):

High RDI → Increased Neural-Emotional Load contribution
Moderate RDI → Adaptive strain zone
Low RDI → Regulated state

THIS FEEDS:

BSI Domains:

- Neural-Emotional Load
- Adaptive Capacity
- Stress Regulation Stability
- Recovery Readiness

This becomes one weighted contributor among:

- Self-report (Art of Wellness)
- Biomarkers
- HRV (if available)
- Sleep data
- Environmental load

It does not stand alone.

It integrates.

What your run produced:

- `results/stage5_model_output.csv`
- `results/stage5_results.json`

That means Stage 5 completed as a **Neural-Autonomic Load Translation pass**, which is exactly the intended purpose of this stage.

The code aligns with these Stage 5 elements:

- Neural-Autonomic Load Translation Engine
- Integration into BSI™ as a **weighted contributor**
- Non-diagnostic mapping logic
- Sympathetic activation patterns (activation / load indicators)
- Parasympathetic engagement (regulation / recovery indicators)
- Rigidity vs flexibility modeling (constraint vs adaptability balance)
- Recovery bandwidth (capacity to return toward baseline)
- Adaptive transition capacity (efficiency of state shifting)
- Model positioning that **does not assign emotional meaning**
- Translation into BSI contribution domains:
    - Neural-Emotional Load
    - Adaptive Capacity
    - Stress Regulation Stability
    - Recovery Readiness
- Outcome framing: **this model does not stand alone and integrates with other contributors** (self-report, biomarkers, HRV, sleep, environmental load)

Your current Stage 5 results indicate that:

- Stage 4 regulation stability outputs were successfully translated into higher-level autonomic regulation constructs
- the model generated composite scores representing:
    - activation / load
    - regulation / recovery
    - flexibility vs rigidity
    - recovery range
    - transition efficiency
- those modeled constructs were then translated into structured BSI contribution domains
- the current session maps into a **regulated-state contribution profile** under the present weighting logic
- the pipeline generated:
    - a structured CSV output for backend modeling
    - a structured JSON output for readable downstream integration
- Stage 5 is functioning correctly as a **translation and contribution layer**, not as a preprocessing, feature extraction, or diagnostic stage

What the numbers mean (critical clarification):

- all Stage 5 outputs are **modeled composite indices on a 0–100 scale**
- they are **dimensionless (no physical units like Hz, seconds, or dB)**
- they represent **relative contribution strength within the model**
- they do **not diagnose emotional or medical states**
- they reflect how the signal maps into **autonomic regulation domains**

Example interpretation of your outputs:

- **Neural-Emotional Load (~18)**
  → Low relative load contribution (lower modeled strain / activation burden)
- **Adaptive Capacity (~73)**
  → Strong ability to shift and adapt across states
- **Stress Regulation Stability (~72)**
  → Stable and coordinated regulation patterns
- **Recovery Readiness (~77)**
  → Strong capacity to return toward baseline

Model positioning (key to your framework):

- This model is **non-diagnostic**
- It does **not assign emotional meaning**
- It focuses on:
  - coordination
  - stability
  - variability
  - adaptability
- It is designed as **one contributor within a broader BSI™ system**, not a standalone decision engine

Outcome framing:

Stage 5 successfully translates signal-derived structure into **interpretable autonomic regulation contributions**, while preserving:

- non-diagnostic integrity
- structural (not emotional) interpretation
- compatibility with multi-input systems

That is the correct role of Stage 5

STAGE 5 CODE CHANGE FOR RSI ALIGNMENT:

What changed from the old BSI-style Stage 5
The biggest differences are:

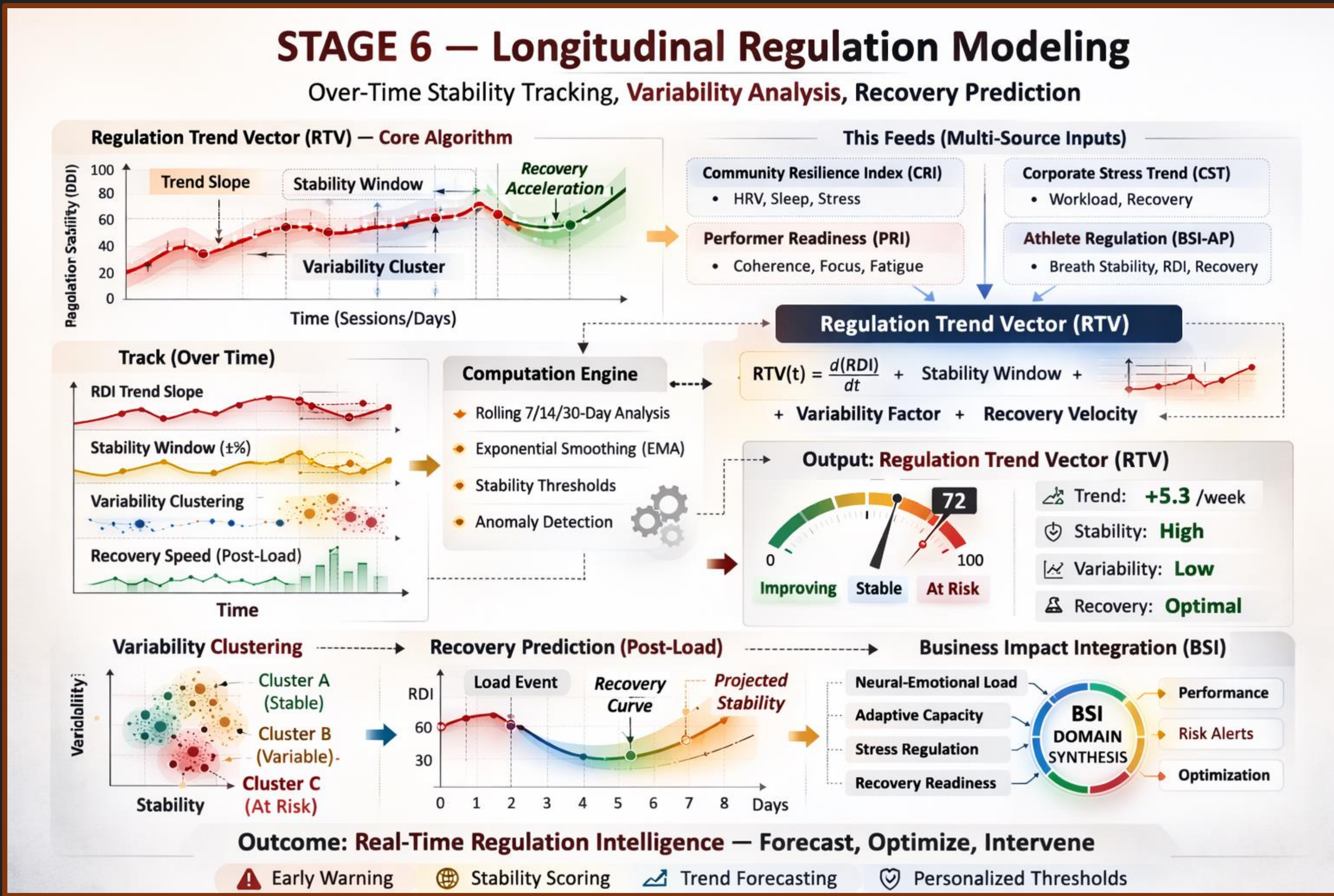
- **same upstream math engine**
- **different final destination labels**
- **different domain mapping**

Old destination:

- BSI-style contributions

New RSI destination:

- **Biological Strain**
- **Neural-Emotional Load**
- **Recovery & Resilience Capacity**
- **Adaptive Capacity**



STAGE 6 — Longitudinal Regulation Modeling

This is where your CTO builds real value.

OVER TIME:

Track:

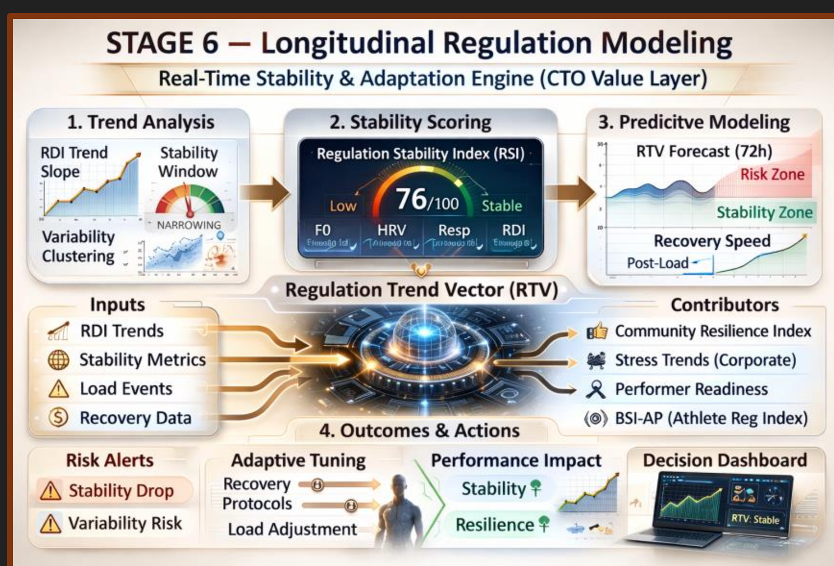
- RDI trend slope
- Stability window narrowing/widening
- Variability clustering
- Recovery speed after load spikes

Create:

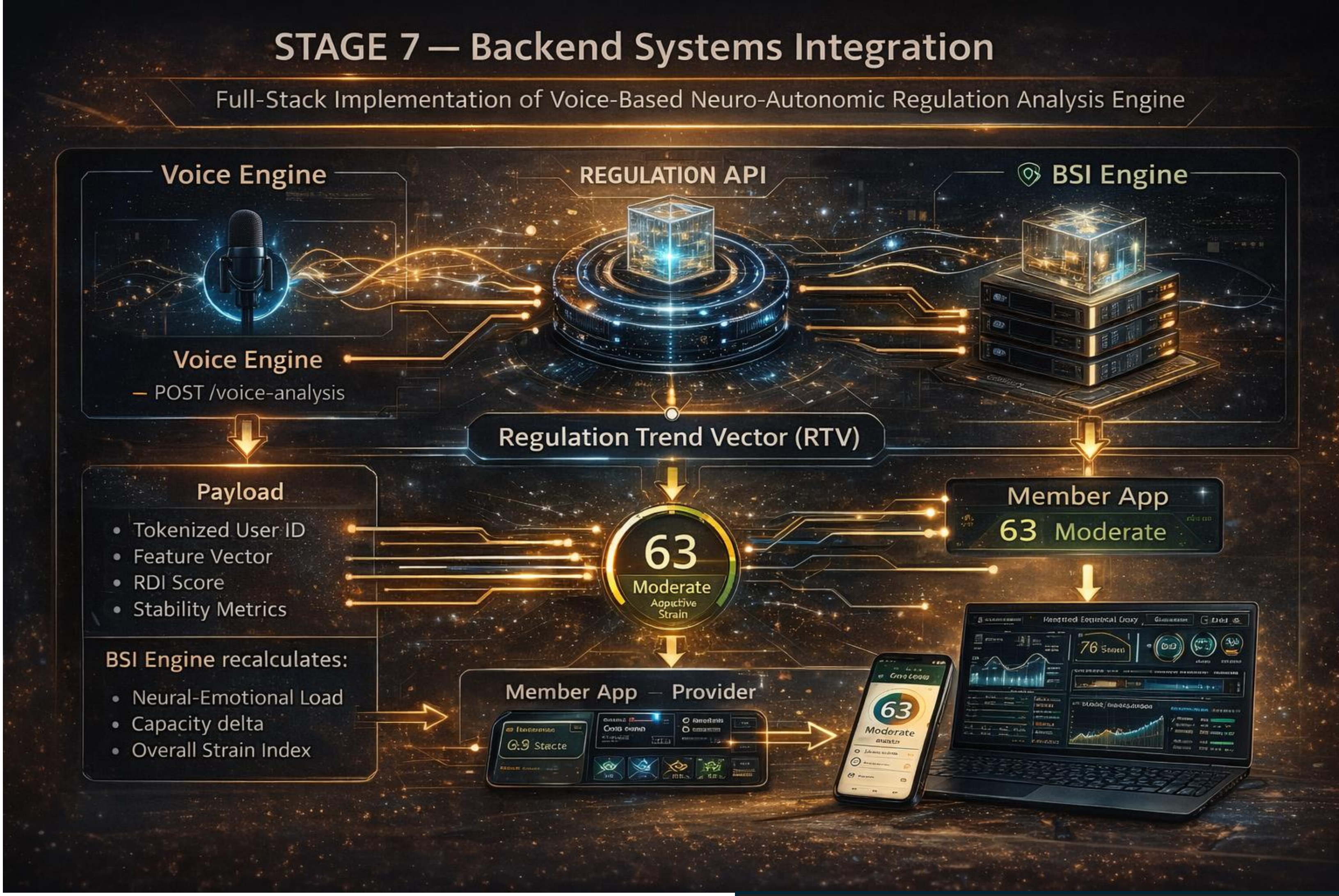
Regulation Trend Vector (RTV)

This feeds:

- Community Resilience Index
- Corporate Stress Trend Analysis
- Performer Readiness Index
- Athlete Regulation Index (BSI-AP)



FIGURES: Longitudinal voice-derived regulation metrics are tracked over time to construct a Regulation Trend Vector (RTV) reflecting stability windows, variability clustering, and recovery velocity following load events. The computation engine applies rolling time windows, threshold modeling, anomaly detection, and smoothing logic to identify adaptive shifts rather than single-session snapshots. The resulting RTV feeds directly into BSI™ domains including Neural-Emotional Load, Adaptive Capacity, Stress Regulation, and Recovery Readiness strengthening predictive modeling across verticals. This transforms isolated regulation signals into time-based intelligence that supports forecasting, optimization, and proactive intervention within the Breathermae ecosystem.



STAGE 7 — Backend Systems Integration

Now we talk architecture.

API Structure

Voice Engine → Regulation API → BSI Engine

POST /voice-analysis

Payload:

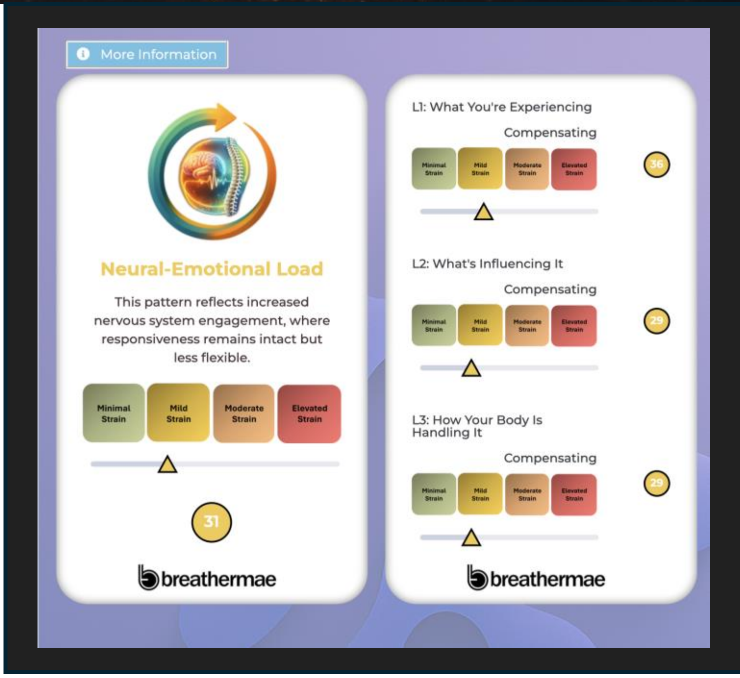
- Tokenized user ID
- Feature vector
- RDI score
- Stability metrics

BSI Engine recalculates:

- Neural-Emotional Load
- Capacity delta
- Overall strain index

Updated dashboard pushed to:

- Member App
- Provider Dashboard
- Corporate Analytics Panel



FIGURES: Standardized voice samples are converted into non-diagnostic regulation metrics including stability, variability, coordination, and load indicators. The metrics are transmitted into the Breathermae Systems Intelligence (BSI™) Engine. Within BSI™, these autonomic signals are weighted alongside Art of Wellness inputs, Key Life Essentials drivers, longitudinal trends, and cross-domain load and recovery data. The combined modeling updates the Neural-Emotional Load Card to reflect adaptive strain, regulation capacity, and system coordination without diagnosing or labeling psychological states. The Voice Engine does not interpret emotion; instead, it models regulation and translates acoustic stability into actionable systems intelligence across the Breathermae platform.

STAGE 8 — Regulatory Safeguards

HARD-CODED CONSTRAINTS:

System must never:

- Output “anxiety detected”
- Output “depression likelihood”
- Output psychiatric terms
- Diagnose

Allowed outputs:

- Regulation Stability
- Variability Index
- Load Indicator
- Coordination Score
- Trend Signal

This keeps it patent-safe and regulatory-safe.

STAGE 9 — Engineering Sequence Roadmap

CTO & CSO CONSIDERATIONS:

1. Build Capture SDK
2. Build Preprocessing Pipeline
3. Validate Feature Extraction Accuracy
4. Build Baseline Modeling Engine
5. Construct RDI Composite Scoring
6. Integrate into BSI Weighting Logic
7. Deploy Longitudinal Modeling
8. Validate across 100–300 pilot samples
9. Refine weighting based on physiologic correlation

STAGE 9 — Engineering Sequence Roadmap

CTO & CSO Strategic Build Priorities

CTO & CSO CONSIDERATIONS:

- Build Capture SDK
- Build Preprocessing Pipeline
- Validate Feature Extraction Accuracy
- Build Baseline Modeling Engine
- Construct RDI Composite Scoring
- Integrate into BSI Weighting Logic
- Deploy Longitudinal Modeling
- Validate across 100–300 pilot samples
- Refine weighting based on physiologic correlation

*No mental health diagnostic terms.

Strategic Relevance to Breathermae

- A non-invasive neural load sensor
- A daily check-in tool
- A corporate resilience tracker
- A performer readiness signal
- PTSD-support regulation monitor (without diagnosing)

IT FITS INSIDE:

- Art of Wellness → subjective
- Voice Engine → autonomic
- BSI → integrative interpretation

Engineering Sequence Roadmap

CTO & CSO CONSIDERATIONS:

1. Build Capture SDK
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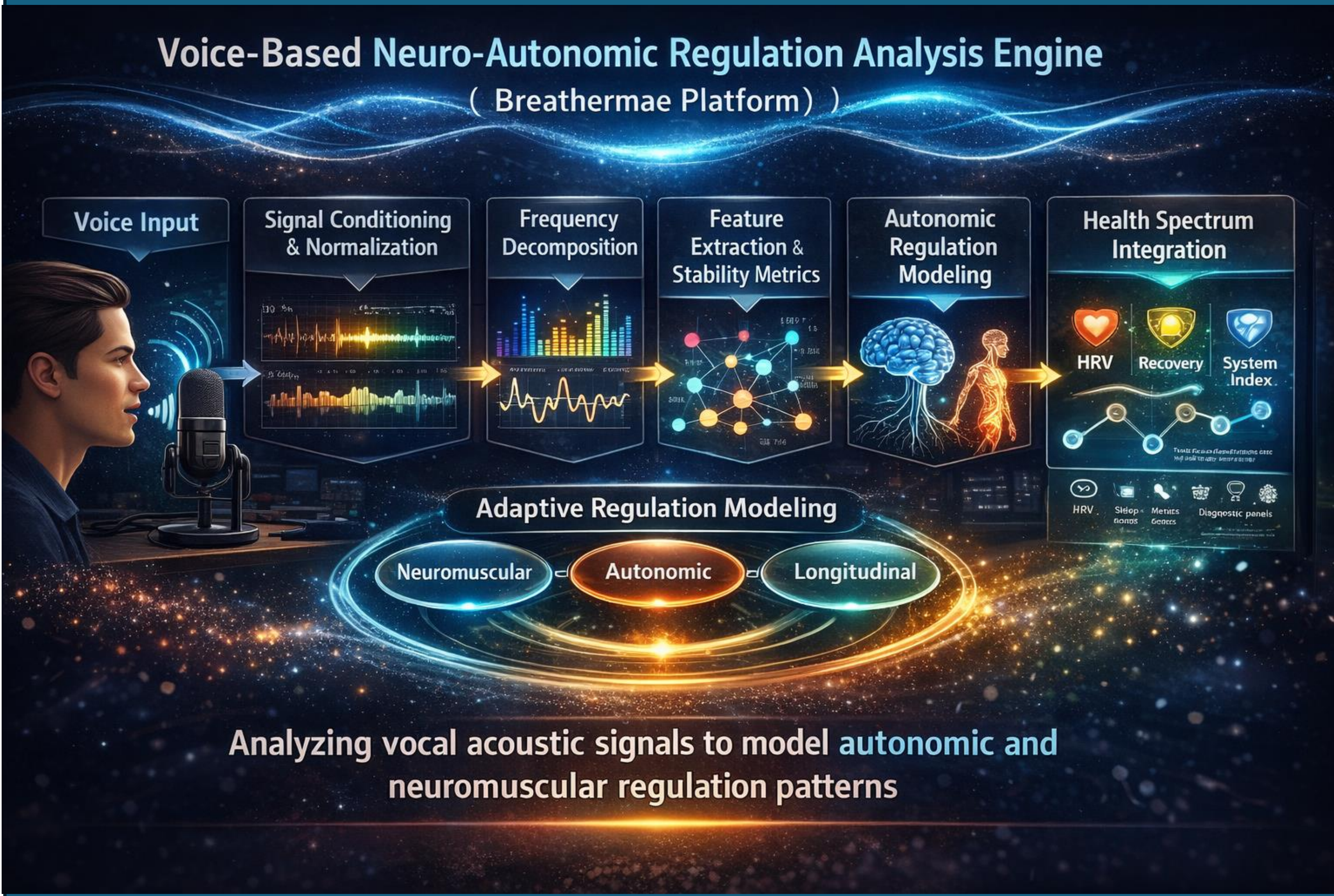
5. Construct RDI Composite Scoring
6. Integrate into BSI Weighting Logic
7. Deploy Longitudinal Modeling
8. Validate across 100–300 pilot s amples

TOGETHER:

- You measure how someone feels
- You measure how someone's autonomic system regulates
- You measure how someone uniquely responds to stress

That is the differentiation



Voice-Based Neuro-Autonomic Regulation Analysis Engine

Patent-Safe System Architecture

Date: February 26, 2026

Voice-Based Neuro-Autonomic Regulation Analysis Engine

Patent-Safe System Architecture

(Breathermae Platform)

SYSTEM OVERVIEW (Top-Level Claim Framing)

The system converts raw vocal signals into interpretable indicators of autonomic regulation, neuromuscular coordination, and neuro-emotional load through multi-stage signal decomposition, pattern stability analysis, and longitudinal state modeling.

The system does not diagnose emotional states or medical conditions and does not rely on fixed frequency-emotion mappings.

SYSTEM BLOCK DIAGRAM (Textual)

[Voice Input Interface]
↓
[Signal Conditioning & Normalization]
↓
[Frequency Decomposition Engine]
↓
[Feature Extraction & Stability Metrics]
↓
[Pattern Distribution & Transition Analysis]
↓
[Autonomic Regulation Modeling Layer]
↓
[Personal Baseline & Trajectory Engine]
↓
[System Integration & Health Spectrum Mapping]

Each block is independently modular and replaceable.

1. Voice Input Interface

Function:
Captures raw acoustic data under controlled or semi-controlled conditions.

Inputs:
• Spoken voice (free speech or prompted)
• Device metadata (microphone type, sampling rate)
• Optional respiratory phase markers

Design Characteristics:
• No linguistic or semantic parsing
• Content-agnostic (language-independent)
• Emphasis on signal integrity, not meaning

Patent-Safe Language:

"An audio acquisition module configured to capture non-semantic vocal acoustic signals representative of neuromuscular and autonomic activity."

2. Signal Conditioning & Normalization Layer

Function:

Standardizes raw input to reduce device and environment variability.

Processes:

• Noise floor reduction
• Dynamic range normalization
• Sampling rate harmonization
• Respiratory phase segmentation (optional but differentiating)

Key Differentiator:

This layer separates phonation from respiration, which most legacy systems ignore.

Patent-Safe Language:

"A preprocessing module configured to normalize acoustic input and isolate phonatory signal components independent of content or language."

3. Frequency Decomposition Engine

Function:

Transforms time-domain audio into frequency-domain representations.

Key Point:

This system does not identify musical notes.

Instead, it:

• Segments energy across a predefined set of normalized frequency bands
• Bands may be logarithmically spaced and octave-normalized
• Number of bands is configurable (e.g., 8–16)

Important:

Any reference to "12 bands" is treated as an implementation option, not a defining claim.

Patent-Safe Language:

"A frequency analysis module configured to decompose acoustic signals into normalized frequency-energy distributions without reference to musical pitch or tonal identity."

4. Feature Extraction & Stability Metrics

Function:

Extracts quantitative descriptors from each frequency band.

Feature Classes:

- Energy Features: mean amplitude, variance
- Temporal Features: onset, sustain, decay
- Stability Features: jitter, shimmer
- Noise Features: harmonic-to-noise ratio
- Complexity Features: entropy, fractal variation

Critical Differentiation:

The system evaluates stability and regulation, not absolute frequency dominance.

Patent-Safe Language:

"A feature extraction module configured to compute temporal, spectral, and stability-based descriptors indicative of neuromuscular and autonomic regulation."

5. Pattern Distribution & Transition Analysis

Function:

Analyzes how extracted features are distributed and how they change over time.

This Includes:

- Relative dominance vs suppression
- Inter-band coherence
- Transition probability between bands
- Repetition and avoidance patterns

Key Insight:

Emotional and autonomic load manifests in pattern rigidity or volatility, not isolated tones.

Patent-Safe Language:

"A pattern analysis module configured to evaluate distribution balance, inter-band coordination, and temporal transition dynamics of extracted vocal features."

6. Autonomic Regulation Modeling Layer

Function:

Interprets pattern outputs in terms of physiological regulation, not emotion labels.

Outputs May Reflect:

- Sympathetic vs parasympathetic dominance
- Regulatory flexibility
- Neuromuscular tension patterns
- Stress adaptation indicators

No Lookup Tables.
No one-to-one mapping of frequency → emotion → organ.

Patent-Safe Language:

"A regulation modeling module configured to infer autonomic and neuromuscular regulation states from multi-feature vocal pattern dynamics."

7. Personal Baseline & Trajectory Engine

Function:

Anchors interpretation to the individual, not the population.

Capabilities:

- Establish personal baseline profiles
- Track directional change over time
- Measure recovery speed and stability
- Detect adaptive vs maladaptive trends

This is a major IP moat.

Patent-Safe Language:

"A longitudinal modeling module configured to establish individual baseline profiles and compute trajectory-based changes in regulation metrics over repeated measurements."

8. System Integration & Health Spectrum Mapping

Function:

Integrates vocal-derived regulation metrics into Breathermae's broader health architecture.

Integration Targets:

- Neural-Emotional Load Spectrum
- Recovery & Resilience Capacity
- System Coordination Index

Correlative Inputs:

- HRV
- Sleep metrics
- Self-reported assessments
- Diagnostic panels

Patent-Safe Language:

"An integration module configured to map regulation indicators into multi-domain health spectra and correlate with complementary physiological and behavioral data sources."

CLAIM-SAFE POSITIONING SUMMARY

What the System IS:

• A regulation pattern analysis engine
• A longitudinal adaptive signal model
• A non-diagnostic neuro-autonomic indicator

What the System IS NOT:

• A digital twin
• A voice-to-emotion detector
• A medical diagnostic device
• A musical frequency analyzer

SINGLE-SENTENCE MASTER CLAIM (Use This Everywhere)

"A system for analyzing vocal acoustic signals to model autonomic and neuromuscular regulation patterns through frequency-based feature extraction, stability analysis, and longitudinal state modeling."

Voice-Based Neuro-Autonomic Regulation Analysis Engine

Implementation Readiness Framework

Date: March 1, 2026

Prototype BOM

A. Voice Capture

Preferred (validation-grade, reduces confounds)

- USB condenser microphone with:
- 44.1 kHz or 48 kHz capture support (48 kHz preferred)
- cardioid pattern preferred
- low self-noise (specs not critical for v1 if consistent)
- Pop filter (mandatory)
- Mic stand / boom arm (mandatory)
- Distance jig (simple):
- 20–30 cm fixed distance spacer (3D printed or rigid card)
- Quiet-room kit (optional but helpful):
- acoustic foam panels or portable isolation shield

Acceptable substitutes

- Smartphone mic only for later generalization testing (not primary validation)
- Any consistent USB mic across all participants in Phase 1/2 is acceptable

B. ANS / HRV Measurement (minimum required)

Preferred

- Chest strap or ECG device that exports RR intervals (not just BPM)
- Must support:
- timestamped RR series
- 5-minute resting capture

Acceptable substitutes

- Finger PPG device with RR export + validated HRV (if RR series is available)
- If no RR export: use device SDK providing RR; avoid “black-box HRV only” for research

C. Optional Recovery & Context Sensors (recommended)

- Sleep wearable (consistent device model across subjects), capturing:
- sleep duration, sleep efficiency, fragmentation, overnight HRV (if available)
- Blood pressure cuff (resting, for context)
- Salivary cortisol kits (optional "deep validation" arm)

D. Computing & Storage

- 1 dev laptop (Mac/Windows/Linux)
- External encrypted drive or encrypted volume for raw audio storage
- Secure local database (SQLite for v1; Postgres optional)

E. Software Stack (v1)

- Python 3.11+
- Audio + signal processing:
- numpy, scipy, soundfile, librosa

Voice stability features:
- parselmouth (Praat bindings) recommended

Modeling:
- scikit-learn, statsmodels

Data:
- pandas, pyarrow (Parquet)

Dashboard (choose one):
- minimal: streamlit
- or notebook + exported HTML reports

Standard Operating Procedure (SOP)

Prototype Build + Testing Procedure

SOP-0: Project Setup & Versioning

Objective: everything reproducible.

Create a repo with:

- /capture/
- /pipeline/
- /qc/
- /analysis/
- /docs/

Every run writes:

- pipeline_version (git commit hash)
- feature_set_version
- scoring_version
- Store raw audio separately from derived features.

SOP-1: Standard Voice Recording Procedure (All Phases)

Environment

- Quiet room (close door, no HVAC fan if possible)
- No music/TV
- Same recording station each time (for Phase 0/1)

Position

- Seated, upright
- Mic at mouth level
- Distance fixed at 25 cm (±2 cm)

Pre-check (30 seconds)

Log in metadata:

- time of day
- posture
- last caffeine (Y/N + hours)
- last exercise (Y/N + hours)
- current illness symptoms (Y/N)
- sleep quality (0–10)

Recording Format

- WAV, mono, 48 kHz, 16-bit (or 24-bit if simple)
- File name includes session_id only (no names)

Voice Tasks (Total 45–60 seconds)

Task A — Neutral Read (20–25 sec)

Use one standardized paragraph for all sessions.

Task B — Sustained phonation (10 sec)

Comfortable “ah” (no strain).

Task C — Free speech (optional 20 sec)

Neutral topic (no emotional recall).

Immediate QC checks

- clipping detected? (Y/N)
- SNR adequate? (Y/N)
- voiced frames duration ≥ 20 sec total? (Y/N)

If failed → re-record once.

SOP-2: HRV Capture Procedure (All Phases)

Resting HRV Standard

- 5 minutes seated, quiet
- No talking, no phone usage
- Start HRV recording → then voice capture immediately after (or vice versa, but be consistent)

Outputs required

- RR intervals time series
- Derived HRV metrics:
- RMSSD, SDNN
- optional: HF power (if feasible)

SOP-3: Phase 0 — Bench Repeatability (Engineering Validation)

Participants: 3–5 internal

Schedule: 10 sessions/person across 3 days

Per session:

1. HRV 5 min
2. Voice protocol

Analysis:

- Compute ICC for RSI/AFM/RecSI
- Identify top QC failure modes

Stop criteria:

- If ICC RSI < 0.65 → fix pipeline/QC before Phase 1

SOP-4: Phase 1 — Convergent Validity (HRV + Surveys)

Participants: 30–60

Schedule: 7 days, 2 sessions/day (AM within 1 hour of waking; PM 2–3 hours before bed)

Each session:

1. HRV 5 min
2. Voice protocol
3. 60-second check-in:
- stress 0–10
- energy 0–10
- mental clarity 0–10

Primary analyses:

- Correlation: RSI vs RMSSD (within-subject + pooled)
- Mixed-effects models controlling for time-of-day + caffeine + sleep quality

SOP-5: Phase 2 — Sensitivity-to-Change (Interventions)

Design: randomized cross-over within-subject, 2–4 arms

Recommended arms:

A) Paced breathing (6 breaths/min for 5 minutes)
B) Cognitive stressor (2–3 min mental arithmetic under time pressure)
C) Quiet rest (5 minutes)
D) Optional: light exercise (2–3 min step test) + 5 min recovery

Per arm:

1. Baseline HRV 5 min + Voice
2. Intervention
3. Post HRV 5 min + Voice

Primary endpoints:

- Directional change in RSI/AFM/RecSI
- Effect sizes (Cohen's d)
- Agreement with HRV shift direction

SOP-6: Quality Control Rules (Hard Gates)

A voice session is invalid if any:

- clipping > 1% samples
- total voiced duration < 20 sec
- SNR below threshold (set during Phase 0 baseline)
- abnormal device mismatch (wrong mic, wrong sample rate)

An HRV session is invalid if:

- RR series missing
- ectopic/outlier correction exceeds threshold (set by engineer)
- motion artifacts flagged by device

Data Schema

A. File Outputs (per session)

- audio_raw/<session_id>.wav
- features/<session_id>.parquet
- results/<session_id>.json
- qc/<session_id>.json

B. Tables (SQLite/Postgres)

Participants

- participant_id (UUID, de-identified)
- cohort (phase0/phase1/phase2)
- sex_at_birth (optional)
- age_range (optional buckets)
- created_at

Sessions

- session_id (UUID)
- participant_id (FK)
- timestamp_local
- phase (0/1/2)
- arm (breathing/stressor/rest/exercise/null)
- posture (sitting/standing)
- device_mic_id
- sample_rate
- pipeline_version
- feature_set_version
- scoring_version

metadata_self_report

- session_id (FK)
- sleep_quality_0_10
- stress_0_10
- energy_0_10
- clarity_0_10
- caffeine_hours (float)
- exercise_hours (float)
- illness_flag (bool)

hrv_raw

- session_id (FK)
- rr_intervals_ms (array/blob or separate RR table)
- rr_timestamps (optional)

hrv_metrics

- session_id (FK)
- rmssd
- sdnn
- hf_power (optional)
- artifact_ratio

voice_features

(store wide table or parquet; recommended parquet for vectors)

- session_id (FK)
- jitter_local
- shimmer_local
- hnr
- spectral_entropy
- spectral_flux_var
- band_energy_vector (12 floats)
- band_coherence_summary (e.g., mean off-diagonal corr)
- transition_matrix_summary (e.g., entropy, loopiness)

Indices

- session_id (FK)
- RSI
- AFM
- RecSI
- PVP_summary (json)

qc_flags

- session_id (FK)
- snr_ok
- clipping_ok
- voiced_seconds
- sample_rate_ok
- hrv_ok
- notes (text)

Success Criteria Table

Internal GO / NO-GO

Phase 0 — Repeatability (Engineering)

Metric	Target	Pass/Fail
RSI ICC	≥ 0.75	PASS
AFM ICC	≥ 0.65 (v1)	PASS
RecSI ICC	≥ 0.65 (v1)	PASS
QC invalid rate	≤ 10%	PASS

Phase 1 — Convergent Validity (HRV)

Metric	Target	Pass/Fail
RSI vs RMSSD correlation (within-subject)	r ≥ 0.30	PASS
Directional agreement vs HRV shifts (AM vs PM)	≥ 65%	PASS
Mixed-effects model RSI~RMSSD (p-value)	$p < 0.05$	PASS
Adds signal beyond surveys (incremental R^2)	+5–10%	PASS

Phase 2 — Sensitivity to Change

Intervention	Expected	Direction	Target
paced breathing	RSI↑	AFM↑	d ≥ 0.40
cognitive stressor	RSI↓	AFM↓	d ≥ 0.40
quiet rest	small	RSI↑	d ≥ 0.20
exercise + recovery	RSI↓	then rebound	detectable rebound in 10–20 min

Safety / Compliance Guardrails

Constraint	Rule
Claims	wellness + regulation only
Outputs	indices + trends, not diagnoses
Data	no transcript storage
Model	baseline-relative, avoid population “normal/abnormal” labels

Standardized Voice Capture Script (Primary)

"I am speaking at a comfortable pace and volume.

My breathing is natural and unforced.

I am reading these words clearly and evenly.

This recording is capturing how my voice is supported by my body in this moment."

Estimated duration: ~20–25 seconds
Tone: neutral, declarative
Emotional content: none
Cognitive load: low
Cross-cultural readability: high

Sustained Phonation Segment (Immediately After)

"Please take a normal breath and gently sustain the sound 'ah' for as long as it feels comfortable."

Duration: 8–12 seconds
Instruction: no pushing, no strain

Optional Free Speech (Secondary / Longitudinal Use)

Only used after baseline is established.

"For the next twenty seconds, please describe a neutral daily routine, such as preparing a meal or getting ready in the morning."

Why This Script Is Scientifically & Strategically Correct

1. It Eliminates Emotional Priming